

Keng Lin

Email: keng.lin@rutgers.edu
WebPage: oohu.github.io/
Scholar: [Keng Lin](#)
INSPIRE: [Keng Lin](#)
ORCID: [0000-0003-4442-8554](#)

EDUCATION & Work Experience

Ph.D. Candidate, Physics	Rutgers University	Jan. 2021 – Dec. 2026
Post-Baccalaureate Researcher	Los Alamos National Laboratory	Jan. 2020 – Jan. 2021
Research Assistant	Columbia University, Nevis Laboratories	Dec. 2018 – Jan. 2020
B.A., Physics, <i>Cum Laude</i>	Columbia University	Dec. 2018

Awards

URA Visiting Scholar Award: \$15,000	Jul. 2024 – Jan. 2025
--	-----------------------

RESEARCH EXPERIENCE

Collaborator @ SBND (2021 – Present)

Advisor: Professor Andrew Mastbaum

- **Analysis:** Led the heavy QCD axion analysis and co-led the millicharged particle analysis. (Publication incoming)
- **Instrumentation:** Installed and commissioned the Cosmic Ray Tagger (CRT) system.
- **Software:** Developed software development across physics analysis, detector commissioning, and systematic uncertainty simulation. [SBNSoftware](#).
- **Example — Detector Monitoring:** Developed the CRT front-end of the web-based detector monitoring system, with an interactive interface for detector status and monitoring. *Demo available upon request.*

Collaborator @ MicroBooNE (2021 – Present)

Advisor: Professor Andrew Mastbaum

- **Analysis:** Led the full heavy QCD axion analysis pipeline, from methodology and implementation through to execution and final analysis. (Publication incoming)
- **Simulation:** Developed LArSoft-based axion signal event generator. [TwoBodyDecayGen](#)
- **Software:** Developed tools (using C++ and ROOT) for MeV-scale physics analyses.

Collaborator @ MicroBooNE (2018 – 2022)

Advisor: Professor Georgia Karagiorgi

- **Analysis:** Validate external-events for the flagship single-photon LEE analysis. [Poster](#)
- **Analysis:** Applied boosted decision trees to isolate electron neutrino signal candidates. [Slides](#)

Collaborator @ MiniBooNE (2020)

Advisors: Dr. William Louis and Dr. Richard Van De Water

- **Analysis:** Performed data-background comparisons and assessed the compatibility of signal hypothesis with observed excesses. [Phys. Rev. D 103, 052002](#).

Research Assistant @ Coherent CAPTAIN-Mills (CCM) Experiment (2020)

Advisors: Dr. William Louis and Dr. Richard Van De Water

- **Hardware:** Decommissioned the CCM120 first-generation detector.

Research Assistant @ Columbia University (2018)

Advisor: Professor Yuri Levin

- **Simulation:** Implemented an event simulator for the collision of a black hole with a hypothetical cosmic string in C.

PUBLICATIONS & CONFERENCE TALKS & POSTERS

Selected Publications:

MiniBooNE Collaboration, "Updated MiniBooNE Neutrino Oscillation Results with Increased Data and New Background Studies", Phys. Rev. D 103,052002

MicroBooNE Collaboration, "The MicroBooNE Single-Photon Low-Energy Excess Search", MICROBOONE-NOTE-1087-PUB

CONFERENCE TALKS & POSTERS

Talks:

Searches for Physics Beyond the Standard Model with the Short-Baseline Near Detector
Lepton-Photon 2025 [Slides](#)

MicroBooNE's Beyond Standard Model Physics Program
NuFact 2024 [Slides](#)

Current and Future Perspectives of MicroBooNE's BSM Program
APS April Meeting 2024 [Slides](#)

Toward a Heavy Axion Selection in MicroBooNE Using the NuMI Beam
APS April Meeting 2023 [Slides](#)

Constraining "Dirt" Backgrounds for the MicroBooNE Single-Photon Low-Energy Excess Search
New Perspectives 2020 [Slides](#)

Study Neutrinos Using the MiniBooNE Detector
Los Alamos National Laboratory Symposium 2020 [Recording](#)

Boosted Decision Tree-Based Single-Electron Selection in MicroBooNE
APS April Meeting 2019 [Slides](#)

Posters:

Event Selection Tools Targeting Single-Photon Events in the Short-Baseline Near Detector
Neutrino 2022 [Poster](#)

Constraining "Dirt" Backgrounds for the MicroBooNE Single-Photon Low-Energy Excess Search
Neutrino 2020 [Poster](#)

Press

New Machine Helps Scientists Detect Mysterious Particles [Article](#)

TEACHING EXPERIENCE

Teaching Assistant @ Rutgers University (2021, 2025)
Advanced General Physics (Fall 2025); *General Physics* (Fall 2025); *General Physics Lab* (Summer 2025, Spring 2021)

Instructor of Materials Science and Nanotechnology @ Columbia University (Summer 2019)

PUBLIC OUTREACH

Science Demonstrator at The Franklin Institute, Philadelphia, PA (2016)

Science Demonstrator at Guangdong Science Center, Guangzhou, China (2012)

Technical Skills

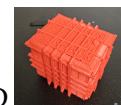
Programming:	C, C++, Python, Mathematica, R, MATLAB, JavaScript, Bash
Physics/Scientific:	ROOT, LArSoft, Geant4, G4BNB, Pythia8, PyHF
Machine Learning:	XGBoost
Data/Analysis:	NumPy, pandas, Matplotlib
Engineering:	Siemens NX, LabVIEW
Development:	Git, CMake, HTML, XML, Markdown, $\LaTeX$
Operating Systems:	Linux, macOS, Windows

SIDE PROJECTS

3D-Printed Detectors: Scaled-down models of the MicroBooNE



and SBND



NeutrinoExperimentsOnEarth: An interactive map of neutrino experiments around the world.
[\[View Map\]](#)